

legoESM: A Differentiable Earth System Model in JAX

Technical Documentation of Equations and Implementations

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Contents

1	Introduction	4
2	Grid Systems	4
2.1	Cubed-Sphere Grid	4
2.1.1	Face-to-Cartesian mapping	4
2.1.2	Cell area	4
2.1.3	Grid spacing	4
2.1.4	Grid angle and wind rotation	4
2.1.5	Halo exchange	5
2.2	Gaussian Grid and Spectral Transforms	5
2.2.1	Grid dimensions	5
2.2.2	Associated Legendre polynomials	5
2.2.3	Spherical harmonic transforms	5
2.2.4	Spectral operators	5
2.2.5	Velocity from vorticity and divergence	6
2.3	Sigma Vertical Coordinate	6
2.3.1	Simmons–Burridge geopotential	6
2.3.2	Sigma-dot	6
2.3.3	Pressure velocity	6
2.4	Height and z^* Coordinates	6
3	Core Operators	7
3.1	Gradient	7
3.2	Divergence (flux form)	7
3.3	Vorticity	7
3.4	Laplacian and Hyperdiffusion	7
4	Finite-Volume Transport	7
4.1	MUSCL Reconstruction (2nd order, halo = 1)	7
4.2	PPM Reconstruction (4th order, halo = 2)	7
4.2.1	Fourth-order edge values	8
4.2.2	Monotonicity constraint	8
4.2.3	Colella–Woodward limiting	8
4.3	WENO5 Reconstruction (5th order, halo = 2)	8
4.3.1	Left-biased reconstruction at $i + \frac{1}{2}$	8
4.3.2	Smoothness indicators	8

4.3.3	Nonlinear weights	8
4.3.4	Right-biased reconstruction	8
4.4	Rusanov Flux	9
4.5	Well-balanced scheme	9
5	Atmospheric Dynamics	9
5.1	Shallow Water Equations	9
5.2	Hydrostatic Primitive Equations (σ -coordinates)	9
5.3	Spectral Primitive Equations (Vorticity–Divergence)	10
5.4	Non-Hydrostatic Compressible Euler Equations	10
5.4.1	Split-explicit time integration	10
5.4.2	Sponge layer	11
6	Atmospheric Physics	11
6.1	Gray Radiation (Frierson 2006)	11
6.1.1	Longwave optical depth	11
6.1.2	Two-stream LW fluxes	11
6.1.3	Shortwave (Beer–Lambert)	11
6.1.4	Heating rate	11
6.2	RRTMGP Correlated- k Radiation	12
6.3	SBM Convection (Frierson 2007)	12
6.3.1	Moist adiabatic lapse rate	12
6.3.2	Saturation mixing ratio (Tetens)	12
6.3.3	CAPE and trigger	12
6.3.4	Enthalpy-conserving relaxation	12
6.3.5	Precipitation	12
6.4	Held–Suarez Forcing	12
6.4.1	Equilibrium temperature	12
6.4.2	Newtonian relaxation	13
6.4.3	Rayleigh friction	13
6.5	Louis (1979) Turbulence	13
6.5.1	Mixing length	13
6.5.2	Richardson number	13
6.5.3	Stability functions	13
6.6	Vertical Diffusion (Implicit Thomas Algorithm)	13
6.7	Surface Fluxes (Bulk Aerodynamic)	14
7	Ocean Dynamics	14
7.1	Boussinesq Hydrostatic Primitive Equations	14
7.1.1	Tracer equations (skew-symmetric)	14
7.1.2	Free-surface equation	14
7.2	Barotropic Solver (Forward–Backward, Semi-Implicit Coriolis)	14
7.3	Equation of State (Wright 1997)	14
7.4	Hydrostatic Pressure and Vertical Velocity	15
8	Time Integration	15
8.1	SSP-RK3 (Shu–Osher 1988)	15
8.2	Split-Explicit Methods	15

9	Land, Ice, and Coupler	15
9.1	Slab Land Model	15
9.1.1	Energy balance	15
9.1.2	Bucket hydrology	15
9.2	Sea-Ice Model	15
9.2.1	Surface temperature	15
9.2.2	Growth and melt	16
9.2.3	Concentration	16
9.3	Tile Coupler	16
10	Machine Learning: Spherical Fourier Neural Operator	16
10.1	Architecture	16
10.2	Spectral convolution	16
11	Conservation Corrections	16
11.1	Atmosphere	16
11.2	Ocean	17
12	Physical Constants	17
13	References	17

Introduction

legoESM is a modular Earth System Model written entirely in JAX, supporting automatic differentiation through all components. The model couples atmospheric, oceanic, land, and sea-ice components on the sphere, with nine dynamical cores and over thirty physics parameterisations.

Design principles.

- Pure-JAX implementation: all components are compatible with `jit`, `grad`, and `vmap`.
- Modular physics: each parameterisation follows a factory pattern returning a callable `physics_fn(state, grid, coord) → Tendancies`.
- Multiple dynamical cores share the same physics interface, enabling interchangeable experiments.
- Neural-network components (SFNO) integrate seamlessly via the same JAX computational graph.

Grid Systems

Cubed-Sphere Grid

Six faces of a cube are mapped to the sphere via **gnomonic (central) projection**. Let $(\alpha_x, \alpha_y) \in [-\pi/4, \pi/4]^2$ be gnomonic coordinates on a face.

Face-to-Cartesian mapping

For face 0 (+x face):

$$(x, y, z) = (1, \tan \alpha_x, \tan \alpha_y), \quad (1)$$

with analogous expressions for the remaining five faces. Points are normalised to the unit sphere: $\hat{\mathbf{r}} = \mathbf{r}/|\mathbf{r}|$, and geographic coordinates follow as $\lambda = \text{atan2}(\hat{y}, \hat{x})$, $\phi = \arcsin(\hat{z})$.

Cell area

The area element on the gnomonic equidistant grid is

$$\Delta A = \frac{R^2 \Delta \alpha^2}{\cos^2 \alpha_x \cos^2 \alpha_y D^3}, \quad D = \sqrt{1 + \tan^2 \alpha_x + \tan^2 \alpha_y}, \quad \Delta \alpha = \frac{\pi}{2n}. \quad (2)$$

Grid spacing

Great-circle distances use the chord-to-arc formula:

$$d = R \cdot 2 \arcsin\left(\frac{|\mathbf{r}_1 - \mathbf{r}_2|}{2}\right). \quad (3)$$

Metric factors $\Delta x_{i,j}$ and $\Delta y_{i,j}$ are computed as two-cell spans for second-order centred differences.

Grid angle and wind rotation

The angle θ between the grid x -axis and geographic east is

$$\theta = \text{atan2}\left(\frac{\partial \phi}{\partial i}, \frac{\partial \lambda}{\partial i} \cos \phi\right). \quad (4)$$

Wind rotation between geographic (u_E, v_N) and grid-aligned (u_g, v_g) :

$$\begin{pmatrix} u_E \\ v_N \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} u_g \\ v_g \end{pmatrix}. \quad (5)$$

Halo exchange

Scalar fields are exchanged directly across the six-face connectivity table. Vector fields are first rotated to geographic components, padded as scalars, then rotated back using the padded grid angle.

Two halo widths are supported. **Halo = 1** (one ghost cell per side) suffices for second-order centred differences and MUSCL reconstruction. **Halo = 2** (two ghost cells per side) is required by higher-order reconstruction stencils (PPM, WENO5). The halo exchange interpolates neighbour-face strips to the correct gnomonic position at each depth, removing the $O(\Delta\alpha)$ mismatch near cube corners.

Gaussian Grid and Spectral Transforms

The spectral dynamical cores use a Gaussian grid with triangular truncation at total wavenumber $n_{\max}$.

Grid dimensions

$$n_{\text{lat}} = \lceil \frac{3}{2}(n_{\max} + 1) \rceil_{\text{even}}, \quad n_{\text{lon}} = 2 n_{\text{lat}}, \quad n_{\text{sh}} = \frac{(n_{\max} + 1)(n_{\max} + 2)}{2}. \quad (6)$$

Latitudes are Gaussian quadrature points $\mu_k = \sin \phi_k$ with weights w_k ; longitudes are equispaced.

Associated Legendre polynomials

Fully normalised (4π) polynomials $P_n^m(\mu)$ satisfy $\int |Y_n^m|^2 dA = 1$. They are computed via the recursion

$$P_m^m = -\sqrt{\frac{2m+1}{2m}} \sqrt{1-\mu^2} P_{m-1}^{m-1}, \quad (7)$$

$$P_n^m = \sqrt{\frac{4n^2-1}{n^2-m^2}} \left(\mu P_{n-1}^m - \sqrt{\frac{(n-1)^2-m^2}{4(n-1)^2-1}} P_{n-2}^m \right). \quad (8)$$

The derivative operator $H_n^m = -(1-\mu^2) dP_n^m/d\mu$ is used for meridional gradients.

Spherical harmonic transforms

Analysis (grid $\rightarrow$ spectral).

$$\hat{f}_{nm} = 2\pi \sum_k w_k P_n^m(\mu_k) \tilde{f}_m(\mu_k), \quad \tilde{f}_m = \frac{1}{n_{\text{lon}}} \text{FFT}[f]_m. \quad (9)$$

Synthesis (spectral $\rightarrow$ grid).

$$f(\phi, \lambda) = \text{iFFT} \left[\sum_{n=m}^{n_{\max}} \hat{f}_{nm} P_n^m(\mu) \right]. \quad (10)$$

Spectral operators

Laplacian eigenvalue: $\nabla^2 \hat{f}_{nm} = -n(n+1)/a^2 \cdot \hat{f}_{nm}$. Inverse Laplacian: $\nabla^{-2} \hat{f}_{nm} = -a^2/[n(n+1)]$ for $n > 0$. Hyperdiffusion: $-\nu[n(n+1)/a^2]^p \hat{f}_{nm}$.

Velocity from vorticity and divergence

Streamfunction $\psi = \nabla^{-2}\zeta$, velocity potential $\chi = \nabla^{-2}\delta$. Velocity reconstruction:

$$u \cos \phi = \frac{\partial \psi}{\partial \theta} + \frac{\partial \chi}{\partial \lambda}, \quad v \cos \phi = \frac{\partial \psi}{\partial \lambda} - \frac{\partial \chi}{\partial \theta}, \quad (11)$$

where θ is colatitude.

Sigma Vertical Coordinate

$$\sigma = p / p_s, \quad \sigma \in [\sigma_{\text{top}}, 1]. \quad (12)$$

Half-level interfaces $\sigma_{k+1/2}$ are uniformly spaced; full levels $\sigma_k = (\sigma_{k-1/2} + \sigma_{k+1/2})/2$. Layer thickness $\Delta\sigma_k = \sigma_{k+1/2} - \sigma_{k-1/2}$.

Simmons–Burridge geopotential

The geopotential at full level k is

$$\Phi_k = \Phi_s + R_d \sum_{k'=k+1}^{N-1} T_{k'} \ln \frac{\sigma_{k'+1/2}}{\sigma_{k'-1/2}} + \alpha_k R_d T_k, \quad (13)$$

where the Simmons–Burridge coefficient is

$$\alpha_k = 1 - \frac{\sigma_{k-1/2}}{\Delta\sigma_k} \ln \frac{\sigma_{k+1/2}}{\sigma_{k-1/2}}. \quad (14)$$

Sigma-dot

The vertical velocity in σ -space is obtained by integrating the continuity equation from the model top ($\dot{\sigma} = 0$):

$$\dot{\sigma}_{k+1/2} = \frac{\sigma_{k+1/2} - \sigma_{\text{top}}}{1 - \sigma_{\text{top}}} D_{\text{tot}} - \sum_{k'=0}^k \nabla \cdot \mathbf{v}_{k'} \Delta\sigma_{k'}, \quad D_{\text{tot}} = \sum_{k'=0}^{N-1} \nabla \cdot \mathbf{v}_{k'} \Delta\sigma_{k'}. \quad (15)$$

Pressure velocity

Since $p = \sigma p_s$, the material pressure tendency is

$$\omega \equiv \frac{dp}{dt} = \sigma \frac{dp_s}{dt} + p_s \dot{\sigma}, \quad (16)$$

where dp_s/dt is the *material* derivative of surface pressure (Eulerian tendency plus advection by the horizontal wind).

Height and z^* Coordinates

The terrain-following z^* coordinate:

$$z^* = H \frac{z - z_s}{H - z_s}, \quad J = \frac{\partial z}{\partial z^*} = \frac{H - z_s}{H}. \quad (17)$$

Physical height: $z(z^*, x, y) = z_s + z^* (H - z_s)/H$.

For the **ocean**, the z^* coordinate has a dynamic Jacobian:

$$J(\eta) = \frac{\eta + H}{H_{\text{max}}}, \quad (18)$$

where η is the free-surface elevation and H the resting depth.

Core Operators

All finite-difference operators use second-order centred differences on the A-grid (collocated) with halo padding on the cubed sphere. The discrete difference operators δ_i and δ_j are defined as centred differences of *cell-centre* values on the padded grid:

$$\delta_i[q]_{i,j} = q_{i+1,j} - q_{i-1,j}, \quad \delta_j[q]_{i,j} = q_{i,j+1} - q_{i,j-1}. \quad (19)$$

All metric-weighted fluxes ($u h_y, v h_x$) are evaluated at cell centres before differencing, consistent with the A-grid stencil.

Gradient

$$\left. \frac{\partial f}{\partial x} \right|_{i,j} = \frac{f_{i+1,j} - f_{i-1,j}}{2 \Delta x_{i,j}}, \quad \left. \frac{\partial f}{\partial y} \right|_{i,j} = \frac{f_{i,j+1} - f_{i,j-1}}{2 \Delta y_{i,j}}. \quad (20)$$

Divergence (flux form)

$$\nabla \cdot (u, v) = \frac{1}{2 A_{i,j}} \left[\delta_i(u h_y) + \delta_j(v h_x) \right], \quad (21)$$

where $h_x = \Delta x/2$, $h_y = \Delta y/2$ are half-edge lengths and A is the cell area.

Vorticity

$$\zeta = \frac{1}{2 A_{i,j}} \left[\delta_i(v h_y) - \delta_j(u h_x) \right]. \quad (22)$$

Laplacian and Hyperdiffusion

$$\nabla^2 f = \nabla \cdot (\nabla f), \quad \text{Hyperdiffusion:} \quad -\nu \nabla^4 f = -\nu \nabla^2 (\nabla^2 f). \quad (23)$$

Finite-Volume Transport

The finite-volume dynamical cores use upwind reconstruction at cell edges coupled with a Rusanov approximate Riemann solver. Three reconstruction methods are available, selected via the `limiter` parameter.

MUSCL Reconstruction (2nd order, halo = 1)

Piecewise-linear reconstruction with slope limiting (van Leer, 1979). At the interface between cells i and $i+1$:

$$q_L = q_i + \frac{1}{2} \phi_i, \quad q_R = q_{i+1} - \frac{1}{2} \phi_{i+1}, \quad (24)$$

where ϕ_i is the limited slope. Two limiters are provided:

- **Minmod**: $\phi = \text{minmod}(\Delta q_+, \Delta q_-)$,
- **MC** (monotonised central): $\phi = \text{minmod}(\text{minmod}(2\Delta q_+, 2\Delta q_-), \frac{1}{2}(\Delta q_+ + \Delta q_-))$.

PPM Reconstruction (4th order, halo = 2)

The Piecewise Parabolic Method (Colella & Woodward, 1984) constructs a monotone parabola in each cell from two edge values and the cell average.

Fourth-order edge values

At the interface $i + \frac{1}{2}$:

$$\hat{q}_{i+1/2} = \frac{7}{12}(q_i + q_{i+1}) - \frac{1}{12}(q_{i-1} + q_{i+2}). \quad (25)$$

Monotonicity constraint

The edge value is clamped to lie between its two adjacent cell averages:

$$\hat{q}_{i+1/2} \leftarrow \text{median}(\hat{q}_{i+1/2}, q_i, q_{i+1}). \quad (26)$$

Colella–Woodward limiting

For each cell i with left edge value $a_L = \hat{q}_{i-1/2}$ and right edge value $a_R = \hat{q}_{i+1/2}$:

1. If $(a_R - q_i)(q_i - a_L) \leq 0$ (local extremum): set $a_L = a_R = q_i$.
2. Define $\delta_m = a_R - a_L$ and $\delta_6 = 6(q_i - \frac{1}{2}(a_L + a_R))$.
 - If $\delta_m \delta_6 > \delta_m^2$: set $a_L = 3q_i - 2a_R$.
 - If $\delta_m \delta_6 < -\delta_m^2$: set $a_R = 3q_i - 2a_L$.

The left and right states at each edge are then $q_L = a_R$ of the cell to the left and $q_R = a_L$ of the cell to the right.

WENO5 Reconstruction (5th order, halo = 2)

The 5th-order Weighted Essentially Non-Oscillatory scheme (Jiang & Shu, 1996) combines three 3rd-order sub-stencil polynomials with nonlinear weights that adapt to local smoothness.

Left-biased reconstruction at $i + \frac{1}{2}$

Using cells $q_{i-2}, \dots, q_{i+2}$:

$$q^{(0)} = \frac{1}{6}(2q_{i-2} - 7q_{i-1} + 11q_i), \quad d_0 = \frac{1}{10}, \quad (27)$$

$$q^{(1)} = \frac{1}{6}(-q_{i-1} + 5q_i + 2q_{i+1}), \quad d_1 = \frac{6}{10}, \quad (28)$$

$$q^{(2)} = \frac{1}{6}(2q_i + 5q_{i+1} - q_{i+2}), \quad d_2 = \frac{3}{10}. \quad (29)$$

Smoothness indicators

$$\beta_0 = \frac{13}{12}(q_{i-2} - 2q_{i-1} + q_i)^2 + \frac{1}{4}(q_{i-2} - 4q_{i-1} + 3q_i)^2, \quad (30)$$

$$\beta_1 = \frac{13}{12}(q_{i-1} - 2q_i + q_{i+1})^2 + \frac{1}{4}(q_{i-1} - q_{i+1})^2, \quad (31)$$

$$\beta_2 = \frac{13}{12}(q_i - 2q_{i+1} + q_{i+2})^2 + \frac{1}{4}(3q_i - 4q_{i+1} + q_{i+2})^2. \quad (32)$$

Nonlinear weights

$$\alpha_k = \frac{d_k}{(\varepsilon + \beta_k)^2}, \quad \omega_k = \frac{\alpha_k}{\sum_j \alpha_j}, \quad q_L = \sum_k \omega_k q^{(k)}, \quad \varepsilon = 10^{-6}. \quad (33)$$

Right-biased reconstruction

The right state q_R at $i + \frac{1}{2}$ is obtained by mirroring the stencil around the interface, using cells $q_{i-1}, \dots, q_{i+3}$ with ideal weights $d_0 = \frac{1}{10}$, $d_1 = \frac{6}{10}$, $d_2 = \frac{3}{10}$ applied to the reversed sub-stencils.

Rusanov Flux

At each edge, the mass flux is

$$F = \frac{1}{2}(h_L u_L + h_R u_R) - \frac{1}{2} \alpha (h_R - h_L), \quad \alpha = \max(|u_L| + c_L, |u_R| + c_R), \quad (34)$$

where $c = \sqrt{g h}$ is the gravity-wave speed (set to zero for pure transport without wave-speed dissipation). The net flux divergence for cell i is

$$-(\nabla \cdot \mathbf{F})_i = -\frac{1}{A_i} \left[(F_x h_y)_{i+1/2} - (F_x h_y)_{i-1/2} + (G_y h_x)_{j+1/2} - (G_y h_x)_{j-1/2} \right], \quad (35)$$

where h_x, h_y are the edge metric factors.

Well-balanced scheme

When topography h_s is present, the scheme reconstructs the free surface $\eta = h + h_s$ instead of h . The fluid depth at each edge is recovered as $h_L = \max(\eta_L - h_{s,\text{edge}}, 0)$, ensuring zero spurious flux for a lake-at-rest state (Audusse et al., 2004).

Atmospheric Dynamics

Shallow Water Equations

In **vector-invariant form** on the cubed sphere:

$$\frac{\partial h}{\partial t} = -\nabla \cdot (h \mathbf{v}), \quad (36)$$

$$\frac{\partial u}{\partial t} = (\zeta + f) v - \frac{\partial B}{\partial x} + D_u, \quad (37)$$

$$\frac{\partial v}{\partial t} = -(\zeta + f) u - \frac{\partial B}{\partial y} + D_v, \quad (38)$$

where the Bernoulli function is $B = \frac{1}{2}(u^2 + v^2) + g(h + h_s)$ and D_u, D_v represent hyperdiffusion.

The mass equation (36) is discretised using the finite-volume Rusanov flux (Section 5) with a configurable reconstruction method: MUSCL (2nd order), PPM (4th order), or WENO5 (5th order), selected via `ShallowWaterConfig.fv_limiter`. Momentum uses centred differences with velocity-only biharmonic hyperdiffusion.

Hydrostatic Primitive Equations (σ -coordinates)

Prognostic variables: u, v (winds), T (temperature), p_s (surface pressure).

$$\frac{\partial p_s}{\partial t} = - \int_0^1 \nabla \cdot (p_s \mathbf{v}) d\sigma, \quad (39)$$

$$\frac{\partial u}{\partial t} = (\zeta + f) v - \frac{\partial B}{\partial x} - R_d T \frac{\partial \ln p_s}{\partial x} + D_u, \quad (40)$$

$$\frac{\partial v}{\partial t} = -(\zeta + f) u - \frac{\partial B}{\partial y} - R_d T \frac{\partial \ln p_s}{\partial y} + D_v, \quad (41)$$

$$\frac{\partial T}{\partial t} = -\nabla \cdot (T \mathbf{v}) + T \nabla \cdot \mathbf{v} - \dot{\sigma} \frac{\partial T}{\partial \sigma} + \kappa T \frac{\omega}{p} + Q, \quad (42)$$

where $B = K + \Phi$, $K = \frac{1}{2}(u^2 + v^2)$, and $\kappa = R_d/c_p$.

Spectral Primitive Equations (Vorticity–Divergence)

On the Gaussian grid, the momentum equations are recast in the vorticity–divergence form (Bourke, 1972):

$$\frac{\partial \hat{\zeta}}{\partial t} = \nabla \times [\widehat{[(\zeta + f)\mathbf{v}]}] + \widehat{\text{vert. adv.}} - \nu \nabla^{2p} \hat{\zeta}, \quad (43)$$

$$\frac{\partial \hat{\delta}}{\partial t} = \nabla \cdot [\widehat{[(\zeta + f)\mathbf{v}]}] - \nabla^2 (K + \Phi + \widehat{R_d T \ln p_s}) + \widehat{\text{vert. adv.}} - \nu \nabla^{2p} \hat{\delta}, \quad (44)$$

$$\frac{\partial \hat{T}}{\partial t} = -\nabla \cdot (\widehat{T\mathbf{v}}) + T \nabla \cdot \mathbf{v} - \dot{\sigma} \widehat{\partial T / \partial \sigma} + \widehat{\kappa T \omega / p}, \quad (45)$$

$$\frac{\partial \widehat{\ln p_s}}{\partial t} = - \int_0^1 \hat{\delta} \, d\sigma. \quad (46)$$

The spectral curl and divergence of flux vectors $(U, V) = (F_u \cos \phi, F_v \cos \phi)$ are

$$\widehat{\text{curl}} = \frac{im}{a} \mathcal{S}[V / \cos^2 \phi] + \frac{1}{a} \mathcal{S}_\mu[U], \quad \widehat{\text{div}} = \frac{im}{a} \mathcal{S}[U / \cos^2 \phi] - \frac{1}{a} \mathcal{S}_\mu[V], \quad (47)$$

where $\mathcal{S}$ and $\mathcal{S}_\mu$ denote the SH analysis with standard and $dP_n^m/d\mu$ Legendre matrices.

Non-Hydrostatic Compressible Euler Equations

In the z^* terrain-following coordinate, the prognostic variables are u, v, w, θ', ρ' (perturbations from a reference state ρ_0, θ_0, π_0).

$$\frac{\partial u}{\partial t} = (\zeta + f) v - \frac{\partial K}{\partial x} - c_p \theta \frac{\partial \pi'}{\partial x} + D_u - r u, \quad (48)$$

$$\frac{\partial v}{\partial t} = -(\zeta + f) u - \frac{\partial K}{\partial y} - c_p \theta \frac{\partial \pi'}{\partial y} + D_v - r v, \quad (49)$$

$$\frac{\partial w}{\partial t} = -\frac{c_p \theta}{J} \frac{\partial \pi'}{\partial z^*} - g \frac{\theta'}{\theta_0} - r w, \quad (50)$$

$$\frac{\partial \theta'}{\partial t} = -\nabla \cdot (\theta \mathbf{v}_h) - \frac{w}{J} \frac{\partial \theta}{\partial z^*} + D_\theta, \quad (51)$$

$$\frac{\partial \rho'}{\partial t} = -\frac{1}{J} \left[\nabla_h \cdot (J \rho \mathbf{v}_h) + \frac{\partial(\rho w)}{\partial z^*} \right], \quad (52)$$

where $r(z)$ is the sponge-layer Rayleigh damping coefficient and the Exner function perturbation is

$$\pi' = \pi_0 \left[\left(1 + \frac{\rho'}{\rho_0} \right) \left(1 + \frac{\theta'}{\theta_0} \right) \right]^{R_d/c_v} - \pi_0. \quad (53)$$

Split-explicit time integration

The equations are integrated with an outer SSP-RK3 scheme for the slow tendencies (advection, Coriolis, horizontal pressure gradient, diffusion) and inner forward–backward acoustic substeps for the fast variables (w, θ', ρ') .

Forward step (w).

$$w^{n+1} = w^n + \Delta t_s \left[-\frac{c_p \theta}{J} \frac{\partial \pi'}{\partial z^*} - g \frac{\theta'}{\theta_0} \right]. \quad (54)$$

Backward step (ρ', θ').

$$\rho'^{n+1} = \rho'^n - \Delta t_s \frac{\partial(\rho w^{n+1})}{\partial z^*}, \quad \theta'^{n+1} = \theta'^n - \Delta t_s \frac{w^{n+1}}{J} \frac{\partial \theta}{\partial z^*}. \quad (55)$$

An optional **semi-implicit acoustic** solve replaces the forward Euler for w with a tridiagonal system to remove the vertical acoustic CFL constraint.

Sponge layer

$$r(z) = r_{\max} \sin^2\left(\frac{\pi}{2} \frac{\max(0, z - z_{\text{bot}})}{z_{\text{top}} - z_{\text{bot}}}\right). \quad (56)$$

Atmospheric Physics

Gray Radiation (Frierson 2006)

Longwave optical depth

$$\tau(\sigma, \phi) = [f_l \sigma + (1 - f_l) \sigma^4] [\tau_e + (\tau_p - \tau_e) \sin^2 \phi] + \tau_{\text{moist}} q_v, \quad (57)$$

with $\tau_e = 7.2$, $\tau_p = 1.8$, $f_l = 0.2$, $\tau_{\text{moist}} = 1150$.

Two-stream LW fluxes

Layer transmittance and emissivity:

$$\mathcal{T}_k = e^{-D \Delta \tau_k}, \quad \varepsilon_k = 1 - \mathcal{T}_k, \quad D = 1.66. \quad (58)$$

Upward sweep from surface ($F_{\text{sfc}}^\uparrow = \varepsilon_s \sigma_{\text{SB}} T_s^4$):

$$F_k^\uparrow = \mathcal{T}_k F_{k+1}^\uparrow + \varepsilon_k \sigma_{\text{SB}} T_k^4. \quad (59)$$

Downward sweep from TOA ($F_0^\downarrow = 0$):

$$F_k^\downarrow = \mathcal{T}_k F_{k-1}^\downarrow + \varepsilon_k \sigma_{\text{SB}} T_k^4. \quad (60)$$

Shortwave (Beer–Lambert)

$$F_{\text{SW}}^\downarrow(k) = \frac{S_0}{4} e^{-D \tau_{\text{SW},0} \sigma_k}, \quad F_{\text{SW}}^\uparrow(k) = \alpha_s F_{\text{SW}}^\downarrow(\text{sfc}) e^{-D \tau_{\text{SW},0} (1 - \sigma_k)}, \quad (61)$$

with $\tau_{\text{SW},0} = 0.22$, $S_0 = 1360 \text{ W m}^{-2}$, $\alpha_s = 0.31$.

Heating rate

Defining $F_{\text{net}}^\uparrow = F^\uparrow - F^\downarrow$ (upward positive) at each interface, the heating rate for layer k is

$$\left. \frac{dT}{dt} \right|_{\text{rad},k} = \frac{g}{c_p} \frac{F_{\text{net},k+1/2}^\uparrow - F_{\text{net},k-1/2}^\uparrow}{p_{k+1/2} - p_{k-1/2}}, \quad (62)$$

where $k + \frac{1}{2}$ is the bottom interface and $k - \frac{1}{2}$ the top (pressure increasing downward). Positive divergence of upward flux warms the layer.

RRTMGP Correlated- k Radiation

The RRTMGP wrapper calls JAX-RRTMGP with correlated- k gaseous optics (128 LW and 112 SW g -points), a two-stream RTE solver, and optional cloud optics. Default trace-gas concentrations: CO₂=415 ppmv, CH₄=1900 ppbv, N₂O=332 ppbv.

SBM Convection (Frierson 2007)

Moist adiabatic lapse rate

$$\Gamma_m = \frac{R_d T}{c_p p} \frac{1 + L_v q_s / (R_d T)}{1 + L_v^2 q_s / (c_p R_v T^2)}, \quad (63)$$

integrated upward via a trapezoidal predictor–corrector.

Saturation mixing ratio (Tetens)

$$e_s = 611.2 \exp\left(\frac{17.67 T_c}{T_c + 243.5}\right), \quad q_s = \frac{\varepsilon e_s}{p - e_s}, \quad \varepsilon = R_d / R_v \approx 0.622. \quad (64)$$

CAPE and trigger

$$\text{CAPE} = R_d \sum_k \max(0, T_{\text{parcel},k} - T_{\text{env},k}) \frac{\Delta p_k}{p_k}. \quad (65)$$

A smooth sigmoid trigger activates convection:

$$\mathcal{F} = \sigma(\kappa_s (\text{CAPE} - \text{CAPE}_c)), \quad \kappa_s = 0.01 \text{ (J/kg)}^{-1}. \quad (66)$$

Enthalpy-conserving relaxation

The reference profile T_{ref} is obtained by adjusting the moist adiabat so that column-integrated moist enthalpy is conserved:

$$\sum_k [c_p (T_{\text{ref},k} - T_k) + L_v (q_{\text{ref},k} - q_{v,k})] \Delta p_k = 0. \quad (67)$$

Newton iteration (2 steps) solves for the temperature offset. Tendencies:

$$\frac{dT}{dt} = \mathcal{F} \frac{T_{\text{ref}} - T}{\tau_c}, \quad \frac{dq_v}{dt} = \mathcal{F} \frac{q_{\text{ref}} - q_v}{\tau_c}, \quad \tau_c = 7200 \text{ s}. \quad (68)$$

Precipitation

$$P = -\frac{1}{g} \sum_k \frac{dq_v}{dt} \Delta p_k. \quad (69)$$

Held–Suarez Forcing

Equilibrium temperature

$$T_{\text{eq}} = \max\left(200, \left[315 - \Delta T_y \sin^2 \phi - \Delta \theta_z \ln(p/p_0) \cos^2 \phi\right] (p/p_0)^\kappa\right), \quad (70)$$

with $\Delta T_y = 60 \text{ K}$, $\Delta \theta_z = 10 \text{ K}$.

Newtonian relaxation

$$\frac{dT}{dt} = -k_T(\sigma, \phi) (T - T_{\text{eq}}), \quad k_T = k_a + (k_s - k_a) \max\left(0, \frac{\sigma - \sigma_b}{1 - \sigma_b}\right) \cos^4 \phi, \quad (71)$$

with $k_a = 1/(40 \text{ day})$, $k_s = 1/(4 \text{ day})$, $\sigma_b = 0.7$.

Rayleigh friction

$$\frac{d\mathbf{v}}{dt} = -k_v(\sigma) \mathbf{v}, \quad k_v = k_f \max\left(0, \frac{\sigma - \sigma_b}{1 - \sigma_b}\right), \quad k_f = 1/(1 \text{ day}). \quad (72)$$

Louis (1979) Turbulence**Mixing length**

$$l = \frac{\kappa_{\text{vK}} z}{1 + \kappa_{\text{vK}} z / l_{\text{max}}}, \quad \kappa_{\text{vK}} = 0.4, \quad l_{\text{max}} = 100 \text{ m}. \quad (73)$$

Richardson number

$$\text{Ri} = \frac{N^2}{S^2}, \quad N^2 = \frac{g}{\theta_v} \frac{\partial \theta_v}{\partial z}, \quad S^2 = \left(\frac{\partial u}{\partial z}\right)^2 + \left(\frac{\partial v}{\partial z}\right)^2. \quad (74)$$

Stability functions

$$f_m(\text{Ri}) = \begin{cases} 1 - \frac{2b \text{ Ri}}{1 + 3bc l^2 |\text{Ri}|^{1/2} / \Delta z^2} & \text{Ri} < 0, \\ \frac{1}{1 + 2b \text{ Ri} / \sqrt{1 + d \text{ Ri}}} & \text{Ri} \geq 0, \end{cases} \quad (75)$$

with $b = c = d = 5$. Eddy diffusivities: $K_m = l^2 S f_m$, $K_h = K_m$.

Vertical Diffusion (Implicit Thomas Algorithm)

The backward-Euler system $-a_k \varphi_{k-1} + b_k \varphi_k - c_k \varphi_{k+1} = d_k$ is solved with the Thomas algorithm.

Tridiagonal coefficients.

$$a_k = \frac{\Delta t K_{k-1/2}}{\rho_k \Delta z_k \Delta z_{k-1/2}}, \quad c_k = \frac{\Delta t K_{k+1/2}}{\rho_k \Delta z_k \Delta z_{k+1/2}}, \quad b_k = 1 + a_k + c_k. \quad (76)$$

Forward sweep.

$$w_k = \frac{a_k}{b'_{k-1}}, \quad b'_k = b_k - w_k c_{k-1}, \quad d'_k = d_k + w_k d'_{k-1}. \quad (77)$$

Back substitution.

$$\varphi_k = \frac{d'_k + c_k \varphi_{k+1}}{b'_k}. \quad (78)$$

Surface Fluxes (Bulk Aerodynamic)

$$\tau_x = -\rho C_D |\mathbf{V}| u, \quad \tau_y = -\rho C_D |\mathbf{V}| v, \quad (79)$$

$$\text{SH} = \rho c_p C_H |\mathbf{V}| (T_s - T_a), \quad (80)$$

$$\text{LH} = \rho L_v C_H |\mathbf{V}| (q_s - q_a), \quad (81)$$

with $C_D = C_H = 1.5 \times 10^{-3}$ and $|\mathbf{V}| = \sqrt{u^2 + v^2 + 10^{-4}}$.

Ocean Dynamics

Boussinesq Hydrostatic Primitive Equations

In vector-invariant form on the cubed sphere:

$$\frac{\partial u}{\partial t} = (\zeta + f) v - \frac{\partial K}{\partial x} - \frac{1}{\rho_0} \frac{\partial p'}{\partial x} + A_h \nabla^2 u + \frac{\partial}{\partial z} \left(A_v \frac{\partial u}{\partial z} \right) - w \frac{\partial u}{\partial z}, \quad (82)$$

$$\frac{\partial v}{\partial t} = -(\zeta + f) u - \frac{\partial K}{\partial y} - \frac{1}{\rho_0} \frac{\partial p'}{\partial y} + A_h \nabla^2 v + \frac{\partial}{\partial z} \left(A_v \frac{\partial v}{\partial z} \right) - w \frac{\partial v}{\partial z}. \quad (83)$$

The Coriolis force is split: depth-mean $f\mathbf{v}$ in the barotropic substeps, baroclinic shear $f\mathbf{v}'$ in the slow tendencies.

Tracer equations (skew-symmetric)

$$\frac{\partial \phi}{\partial t} = \frac{1}{2} \left[-\mathbf{v} \cdot \nabla \phi - \nabla \cdot (\phi \mathbf{v}) \right] - w \frac{\partial \phi}{\partial z} + K_h \nabla^2 \phi + \frac{\partial}{\partial z} \left(K_v \frac{\partial \phi}{\partial z} \right). \quad (84)$$

Free-surface equation

$$\frac{\partial \eta}{\partial t} = - \sum_k \nabla \cdot (h_k \mathbf{v}_k). \quad (85)$$

Barotropic Solver (Forward–Backward, Semi-Implicit Coriolis)

The barotropic system is solved with split-explicit forward–backward substeps:

$$\frac{\partial \eta}{\partial t} = -\nabla \cdot (H \bar{\mathbf{V}}), \quad \frac{\partial \bar{\mathbf{V}}}{\partial t} = f \hat{\mathbf{k}} \times \bar{\mathbf{V}} - g \nabla \eta + \mathbf{F}_{\text{slow}}. \quad (86)$$

Semi-implicit Coriolis. With $\alpha = \frac{1}{2} f \Delta t_s$:

$$\bar{U}^{n+1} = \frac{R_u + \alpha R_v}{1 + \alpha^2}, \quad \bar{V}^{n+1} = \frac{R_v - \alpha R_u}{1 + \alpha^2}, \quad (87)$$

where $R_u = \bar{U}^n + \alpha \bar{V}^n - \Delta t_s g \partial \eta / \partial x$ and $R_v = \bar{V}^n - \alpha \bar{U}^n - \Delta t_s g \partial \eta / \partial y$.

Equation of State (Wright 1997)

$$\rho = \frac{p + p_0(T, S)}{\lambda(T, S) + \alpha_0(T, S) (p + p_0)}, \quad (88)$$

where

$$\alpha_0 = a_0 + a_1 T + a_2 S, \quad (89)$$

$$p_0 = (b_0 + b_4 S) + T(b_1 + T(b_2 + b_3 T) + b_5 S), \quad (90)$$

$$\lambda = (c_0 + c_4 S) + T(c_1 + T(c_2 + c_3 T) + c_5 S). \quad (91)$$

Coefficients follow the MOM6 implementation. The density perturbation is $\rho' = \rho(T, S, p) - \rho_0$.

Hydrostatic Pressure and Vertical Velocity

Baroclinic pressure by downward integration:

$$p'_k = \rho_0 g \eta + \sum_{k'=0}^{k-1} \rho'_{k'} g \Delta z_{k'} + \frac{1}{2} \rho'_k g \Delta z_k. \quad (92)$$

Vertical velocity diagnosed bottom-up from $w(\text{floor}) = 0$:

$$w_{k-1/2} = - \sum_{k'=k}^{N-1} \nabla \cdot (h_{k'} \mathbf{v}_{k'}). \quad (93)$$

Time Integration

SSP-RK3 (Shu–Osher 1988)

The three-stage strong-stability-preserving Runge–Kutta method:

$$\mathbf{u}^{(1)} = \mathbf{u}^n + \Delta t F(\mathbf{u}^n), \quad (94)$$

$$\mathbf{u}^{(2)} = \frac{3}{4} \mathbf{u}^n + \frac{1}{4} \mathbf{u}^{(1)} + \frac{1}{4} \Delta t F(\mathbf{u}^{(1)}), \quad (95)$$

$$\mathbf{u}^{n+1} = \frac{1}{3} \mathbf{u}^n + \frac{2}{3} \mathbf{u}^{(2)} + \frac{2}{3} \Delta t F(\mathbf{u}^{(2)}). \quad (96)$$

SSP-RK(4,3) and SSP-RK(5,4) variants are also available.

Split-Explicit Methods

Each outer RK stage computes slow tendencies once, updates u, v and tracers, then runs N_s forward–backward acoustic substeps on the fast variables (w, θ', ρ') with substep size $\Delta t_s = \Delta t / N_s$.

Land, Ice, and Coupler

Slab Land Model

Energy balance

$$C_s d_s \frac{dT_s}{dt} = (1 - \alpha_l) SW^\downarrow + \varepsilon_l (LW^\downarrow - \sigma_{\text{SB}} T_s^4) - SH - LH. \quad (97)$$

Bucket hydrology

$$\frac{dW}{dt} = P - E, \quad W \in [0, W_{\text{max}}]. \quad (98)$$

Moisture availability $\beta = \beta_{\text{min}} + (1 - \beta_{\text{min}}) W / W_{\text{max}}$ modulates evaporation via $q_{\text{sfc}} = \beta q_s(T_s)$.

Sea-Ice Model

Surface temperature

$$\rho_i c_i \frac{h_{\text{eff}}}{2} \frac{dT_i}{dt} = Q_{\text{sfc}} - \frac{k_i (T_f - T_i)}{h_{\text{eff}}}, \quad h_{\text{eff}} = \max(h_i, h_{\text{min}}). \quad (99)$$

Growth and melt

$$\frac{dh_i}{dt} = \frac{F_{\text{cond}} - F_{\text{ocean}}}{\rho_i L_f}. \quad (100)$$

Concentration

$$\frac{dc}{dt} = \begin{cases} (dh/dt)(1-c)/h_{\text{new}} & \text{growing,} \\ (dh/dt) c/h_{\text{eff}} & \text{melting.} \end{cases} \quad (101)$$

Tile Coupler

Surface fluxes are area-weighted blends over ocean, land, sea-ice, and lake tiles:

$$X_{\text{blend}} = f_{\text{ocn}} X_{\text{ocn}} + f_{\text{land}} X_{\text{land}} + f_{\text{ice}} X_{\text{ice}} + f_{\text{lake}} X_{\text{lake}}. \quad (102)$$

Fluxes are time-averaged over each coupling window before being passed to the atmosphere.

Machine Learning: Spherical Fourier Neural Operator

The SFNO (ACE/FourCastNet v2 style) operates on the Gaussian grid.

Architecture

1. **Encoder:** Pointwise linear $n_{\text{in}} \rightarrow n_{\text{emb}}$.
2. **N processor blocks**, each containing:
 - Layer normalisation,
 - SH analysis $\rightarrow$ learned spectral convolution $\rightarrow$ SH synthesis,
 - GELU activation,
 - Pointwise MLP (two layers with expansion factor),
 - Skip connection.
3. **Decoder:** Pointwise linear $n_{\text{emb}} \rightarrow n_{\text{out}}$.
4. **Big-skip residual:** output = input + decoder($\dots$).

Spectral convolution

In the SH coefficient space:

$$\hat{f}_{nm}^{\text{out}} = W_{nm} \hat{f}_{nm}^{\text{in}}, \quad (103)$$

where $W_{nm} \in \mathbb{R}^{n_{\text{emb}} \times n_{\text{emb}}}$ is a learnable weight matrix per wavenumber pair.

Conservation Corrections

Atmosphere

Dry-air mass is corrected by a uniform adjustment to p_s that restores the area-weighted global integral:

$$\Delta \bar{p} = \frac{\sum_i A_i p_{s,i}^{\text{new}} - \sum_i A_i p_{s,i}^{\text{old}}}{\sum_i A_i}, \quad p_{s,i}^{\text{corr}} = p_{s,i}^{\text{new}} - \Delta \bar{p}, \quad (104)$$

so that $\sum_i A_i p_{s,i}^{\text{corr}} = \sum_i A_i p_{s,i}^{\text{old}}$ (global dry mass conserved exactly). Column moisture is rescaled: $q^{\text{corr}} = q^{\text{new}} \cdot Q_{\text{old}}/Q_{\text{new}}$.

Ocean

Volume conservation: $\eta^{\text{corr}} = \eta^{\text{new}} + (\sum \eta_{\text{old}} A - \sum \eta_{\text{new}} A) / \sum A$.

Heat and salt conservation: uniform additive correction per unit volume.

Physical Constants

Symbol	Name	Value	Unit
g	Gravitational acceleration	9.80616	m s^{-2}
R_d	Dry-air gas constant	287.04	$\text{J kg}^{-1} \text{K}^{-1}$
c_p	Dry-air heat capacity	1005.0	$\text{J kg}^{-1} \text{K}^{-1}$
R_v	Water-vapour gas constant	461.5	$\text{J kg}^{-1} \text{K}^{-1}$
L_v	Latent heat of vapourisation	2.501×10^6	J kg^{-1}
κ	R_d/c_p	0.285	—
p_0	Reference pressure	10^5	Pa
σ_{SB}	Stefan–Boltzmann constant	5.67×10^{-8}	$\text{W m}^{-2} \text{K}^{-4}$
κ_{vK}	von Kármán constant	0.4	—
Ω	Earth rotation rate	7.292×10^{-5}	rad s^{-1}
$R_{\oplus}$	Earth radius	6.371×10^6	m
ρ_0	Ocean reference density	1025	kg m^{-3}

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